LIBBY — ACCESS GUIDE

Access guide — gist-sdh-multifocal-resected-m1-k4n8

How to access each therapy: trial contacts + manufacturer medical-info

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Access guide — gist-sdh-multifocal-resected-m1-k4n8

How a patient or treating team could practically access each intervention in this case's dossier. Trial recruitment contacts and manufacturer medical-information lines are captured for direct outreach. The number in the first column of the summary table is the entry's identifier in the per-intervention sections below — use it to find the deep section quickly. Information ages — each row carries a Verified date; re-screen before relying on a specific contact or trial slot.

Summary

#	Intervention	Modality	Access status	Regulatory	Recommended first action
1	active surveillance (NED post-R0 dSDH-GIST)	other	Standard of care	n/a — surveillance pathway	Schedule cross-sectional imaging (CT abdomen/pelvis or whole-body MRI) every 3-6 months for the first 2-3 years post-resection, then space out per response.
2	alpelisib (Piqray) — PI3Kα inhibitor for PIK3CA R93W (non-canonical)	small_molecule	Off-label use	FDA-approved (PIK3CA-mutated HR+/HER2- advanced breast cancer in combination with fulvestrant 2019; PIK3CA-related overgrowth spectrum 2022). EMA-approved.	Confirm PIK3CA R93W on tissue NGS before pursuing this path (currently ctDNA-only).
3	belzutifan (Welireg) — HIF-2α inhibitor for dSDH-GIST contingency	small_molecule	Off-label use + Clinical trial	FDA-approved (VHL-associated RCC/CNS 2021; advanced RCC after PD-1/PD-L1 + VEGF-TKI 2023; PPGL 2025). EMA-approved (2025).	Resolve the SDHA/SDHB IHC discrepancy (retained SDHB staining is unusual given biallelic SDHA inactivation) — sponsor will not accept dSDH-GIST labeling without it.
4	cobimetinib (Cotellic) — MEK inhibitor (MEGALiT arm) for MAP2K1 P124S	small_molecule	Off-label use + Clinical trial	FDA-approved (BRAF V600E/K melanoma in combination with vemurafenib; histiocytic neoplasms monotherapy 2022). EMA-approved.	Default to trametinib for MAP2K1 P124S off-label consideration unless a cobimetinib-specific MAP2K1 trial opens.

5	imatinib (Gleevec) — adjuvant or first-line	small_molecule	Off-label use	FDA-approved (KIT-positive GIST, adjuvant and CML and several other indications). EMA-approved.	Document the explicit decision against adjuvant imatinib in the chart citing Boikos 2016 (PMID 27011036) and the ESMO-EURACAN-GENTURIS guideline.
6	inavolisib (Itovebi) — selective PI3Kα inhibitor/degrader for PIK3CA R93W	small_molecule	Off-label use	FDA-approved (HR+/HER2- PIK3CA-mutated advanced breast cancer with palbociclib + fulvestrant 2024). EMA-approved (2025).	Tissue NGS confirmation of PIK3CA R93W (currently ctDNA-only) is the gate.
7	olaparib + temozolomide combination — PARP + alkylator for dSDH-GIST contingency	small_molecule	Off-label use	Both components individually FDA- and EMA-approved (olaparib: BRCA1/2-mutated cancer, HRD-positive ovarian maintenance; temozolomide: glioblastoma, anaplastic astrocytoma, others). Combination is not approved.	Reserve for late-line use after sunitinib/regorafenib/ripretinib and after temozolomide and pemigatinib trial paths are exhausted.
8	pemigatinib (Pemazyre) — FGFR inhibitor for dSDH-GIST contingency	small_molecule	Off-label use + Clinical trial	FDA-approved (cholangiocarcinoma with FGFR2 2020; myeloid/lymphoid neoplasm with FGFR1 rearrangement 2022). EMA-approved	Confirm SDH-deficient phenotype with SDHA/SDHB IHC re-stain and germline SDHA resolution.
9	regorafenib (Stivarga) — 3L TKI for dSDH-GIST contingency	small_molecule	Off-label use	FDA-approved (advanced GIST after imatinib + sunitinib; metastatic CRC; HCC after sorafenib). EMA-approved.	On sunitinib progression, start regorafenib 160 mg PO daily 3 weeks on / 1 week off.

10	ripretinib (Qinlock) — 4L+ TKI for dSDH-GIST contingency	small_molecule	Off-label use	FDA-approved (advanced GIST after ≥3 prior TKIs including imatinib). EMA-approved.	Document prior imatinib, sunitinib, regorafenib lines in the chart before submitting prior auth (label requires ≥3).
11	sunitinib (Sutent) — 2L TKI for dSDH-GIST contingency	small_molecule	Off-label use	FDA-approved (advanced GIST after imatinib failure/intolerance; RCC; pNET; adjuvant RCC). EMA-approved.	At first measurable recurrence, start sunitinib 50 mg PO daily 4 weeks on / 2 weeks off. Confirm dosing with the prescribing oncologist; some dSDH-GIST patients tolerate 37.5 mg continuous better.
12	temozolomide (Temodar) — targeted off-label or trial-based for dSDH-GIST	small_molecule	Off-label use + Clinical trial	FDA-approved (newly diagnosed glioblastoma with radiotherapy and as maintenance; anaplastic astrocytoma; refractory anaplastic astrocytoma — also approved 2024 for additional indications under Project Renewal). EMA-approved. Widely available as generic.	At first recurrence with measurable disease, order MGMT promoter methylation on tumor tissue (or re-cut from the resection block) — this is the load-bearing biomarker for off-label approval and trial enrollment.
13	trametinib (Mekinist) — MEK inhibitor for MAP2K1 P124S	small_molecule	Off-label use + Clinical trial	FDA-approved (BRAF V600E/K melanoma in combination with dabrafenib; BRAF V600E NSCLC; BRAF V600E anaplastic thyroid; BRAF V600E pediatric low-grade glioma). EMA-approved.	Order FoundationOne CDx tissue NGS (or paired tumor + normal) to confirm MAP2K1 P124S as clonal in tumor rather than ctDNA noise.
14	vandetanib (Caprelsa) — VEGFR/RET/EGFR TKI	small_molecule	Off-label use	FDA-approved (Caprelsa) for symptomatic or progressive medullary thyroid cancer that is unresectable, locally advanced, or metastatic. EMA-approved. FDA removed the Caprelsa REMS in 2025.	Do not pursue off-label vandetanib for this patient; the dSDH-GIST trial was a hard negative.

15	NCI Rare Tumor Natural History Study (registry + specialty referral)	other	Clinical trial	n/a — observational registry, no investigational drug.	Email Barbara Thomas (barbara.thomas2@nih.gov) or call 240-858-3633 to start enrollment.
16	RLY-2608 (zovegalisib) — mutant-selective PI3Kα inhibitor for PIK3CA R93W	small_molecule	Clinical trial	Investigational; no approval anywhere. Phase 1 in advanced solid tumors, advancing toward later-phase study in PIK3CA-mutant breast cancer.	Order tissue NGS (FoundationOne CDx or paired tumor/normal) to confirm PIK3CA R93W is clonal in tumor rather than ctDNA noise.
17	olverembatinib (HQP1351) — multikinase inhibitor for dSDH-GIST contingency	small_molecule	Clinical trial + Expanded access program + Not yet accessible	NMPA-approved in China (TKI-resistant T315I chronic- or accelerated-phase CML), and used in China for Ph+ ALL per CSCO guidance. Investigational for GIST everywhere; no FDA or EMA approval for any indication.	Treat olverembatinib as a watch item, not an actionable option, while the patient is NED in the US.
18	rogaratinib (BAY 1163877) — pan-FGFR inhibitor for dSDH-GIST	small_molecule	Clinical trial	Investigational; no FDA or EMA approval. Bayer has stepped back from the broader rogaratinib program, and the surviving study is the NCI sarcoma / dSDH-GIST phase 2.	Use this row as the evidence anchor for the FGFR-inhibitor rationale, not as an enrollment route, since NCT04595747 is closed to new patients.
19	tersolisib (STX-478) — selective PI3Kα inhibitor for PIK3CA R93W	small_molecule	Clinical trial	Investigational; no regulatory approval anywhere. IND-cleared phase 1/2.	Order FoundationOne CDx tissue NGS to confirm PIK3CA R93W as clonal in tumor (currently ctDNA-only).
20	Comprehensive Multinational GIST Registry (observational)	other	Not yet accessible	Not applicable; observational registry with no investigational drug.	Treat as a bridge resource to check back on once it opens recruitment.
21	Wild-Type GIST Oncogenetic Panel Registry (observational)	other	Not yet accessible	Not applicable; observational genomic-profiling registry with no investigational drug.	Do not plan around this registry; it is closed to accrual and based in Taiwan.

22	DFF332 — next-generation HIF-2α inhibitor	small_molecule	Unavailable	Investigational; never approved. Phase 1 terminated Feb 2026 for business reasons.	Mark next-generation HIF-2α inhibitors as off the table beyond belzutifan.
23	guadecitabine (SGI-110) — DNA hypomethylating agent	small_molecule	Unavailable	Investigational; never approved. Phase 3 trials in AML and MDS failed primary endpoints; development discontinued.	Mark hypomethylating-agent strategy as off the table for this patient.
24	linsitinib (OSI-906) — IGF-1R/IR inhibitor	small_molecule	Unavailable	Investigational; never approved. Astellas terminated development after multiple negative trials (adrenocortical carcinoma, GIST).	Mark IGF-1R inhibition as off the table.
25	telaglenastat (CB-839) — glutaminase inhibitor	small_molecule	Unavailable	Investigational; never approved. Development discontinued and the asset divested.	Mark glutaminase inhibition as off the table for this patient.

Standard of care (1)

1. active surveillance (NED post-R0 dSDH-GIST)

Aliases: active surveillance, watch-and-wait, imaging surveillance

Modality: other · **Regulatory:** n/a — surveillance pathway · **Guidelines:** ESMO-EURACAN-GENTURIS 2022 and NCCN GIST guidelines both support surveillance over adjuvant TKI in SDH-deficient GIST given the near-zero imatinib response rate and indolent natural history. · **Geographic scope:** Anywhere with access to standard oncology surveillance imaging. NIH registry enrollment requires US travel to Bethesda for the in-person visits. · **Verified:** 2026-06-26

This is the recommended pathway right now. The patient is NED after R0 resection of a multifocal SDH-deficient GIST, and the published natural history (Mei 2018, NIH 76-pt cohort: median EFS ~2.5 years, ~71% eventually recur) argues for structured surveillance rather than adjuvant imatinib. The ESMO-EURACAN-GENTURIS guideline is explicit on avoiding adjuvant imatinib in this molecular subset.

Next steps

1. Schedule cross-sectional imaging (CT abdomen/pelvis or whole-body MRI) every 3-6 months for the first 2-3 years post-resection, then space out per response.
2. Refer to genetic counseling for the SDHA L306P germline-vs-somatic question and the POLE R1679C germline call — needed for cascade testing and for paraganglioma screening.
3. Add baseline plasma/urine metanephrines + neck-to-pelvis MRI for PGL screening, repeat every 1-2 years

given the Carney-Stratakis risk.
4. Enroll in the NCI Rare Tumor Natural History Study (NCT03739827) for specialty follow-up and biospecimen banking.
5. Document the recurrence-contingency plan now: if relapse, first-line is sunitinib, not imatinib.

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT03739827	n/a	recruiting	yes	Barbara J Thomas, R.N.; barbara.thomas2@nih.gov; (240) 858-3633

Manufacturer / sponsor contact

- **Notes:** Surveillance is a clinic-level pathway with no manufacturer. Reference centers for dSDH-GIST: NIH Pediatric & wt-GIST Clinic (Bethesda), DFCI (Suzanne George), MSKCC (Ping Chi), Royal Marsden (Robin Jones), Lyon Centre Léon Bérard (Jean-Yves Blay).

Payer / coverage: Surveillance imaging (cross-sectional CT or whole-body MRI) is covered under standard oncologic follow-up. Whole-body MRI for SDH-mutation carriers may need prior auth depending on payer; the Mei/Boikos 2018 review and ESMO PPGL screening guidance are the citations to use.

Notes: The single most consequential 'access path' for this patient today is the decision not to start adjuvant imatinib. The systemic-therapy rows below are all contingency plans for first recurrence.

Off-label use (13)

2. alpelisib (Piqray) — PI3Kα inhibitor for PIK3CA R93W (non-canonical)

Aliases: alpelisib, BYL719, Piqray

Modality: small_molecule · **Regulatory:** FDA-approved (PIK3CA-mutated HR+/HER2- advanced breast cancer in combination with fulvestrant 2019; PIK3CA-related overgrowth spectrum 2022). EMA-approved. · **Guidelines:** NCCN-supported only for HR+/HER2- mBC with canonical PIK3CA hotspots (E542K, E545K/A/D/G, H1047R/Y/L, C420R) per the therascreen companion diagnostic. Not on NCCN GIST guidelines. R93W is not on the validated label hotspot list. · **Geographic scope:** US + EU + AU + JP — approved in major markets for HR+/HER2- mBC indication. · **Verified:** 2026-06-26

Approved drug, but the off-label case for R93W is shaky. Burke 2012 biochemistry classifies ABD-domain variants like R88Q/R93 as weakly activating compared with the canonical helical/kinase hotspots, and there is no histology rationale for PI3Kα blockade in GIST. The tersolisib trial is the better-targeted experiment if the variant is the priority; off-label alpelisib should sit in reserve.

Next steps

1. Confirm PIK3CA R93W on tissue NGS before pursuing this path (currently ctDNA-only).

2. Default to the tersolisib trial (NCT05768139) if PI3Kα inhibition is on the table — selective inhibitor with a better therapeutic window than alpelisib.
3. Off-label alpelisib only after PI3Kα trial path is exhausted; even then expect a prior-auth fight and weigh the hyperglycemia/stomatitis profile against the activity expected from a weak ABD-domain variant.

Manufacturer / sponsor contact

- **Company:** Novartis Pharmaceuticals Corporation
- **Med info phone:**1-833-474-7729 (1-833-4-PIQRAY)
- **Product info:**https://www.us.piqray.com/
- **Notes:** Novartis Patient Support: 1-800-282-7630 Mon-Fri 9am-5pm ET. Novartis Oncology Universal Co-pay: 1-877-577-7756.

Payer / coverage: Off-label for GIST AND for non-canonical PIK3CA variants. Companion diagnostic (therascreen PIK3CA RGQ PCR) does not include R93W. Approximate WAC ~$17,000/month. Payers will likely deny on the non-canonical variant alone; hyperglycemia and stomatitis toxicity profile also makes the risk-benefit conversation hard given the patient is currently NED.

Notes: Listed because the user asked. Realistically a back-of-the-queue option for this patient.

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3. belzutifan (Welireg) — HIF-2α inhibitor for dSDH-GIST contingency

Aliases: belzutifan, MK-6482, PT2977, Welireg

Modality: small_molecule · **Regulatory:** FDA-approved (VHL-associated RCC/CNS hemangioblastoma/pNET 2021; advanced RCC after PD-1/PD-L1 + VEGF-TKI 2023; PPGL 2025). EMA-approved (2025). · **Guidelines:** Not on NCCN GIST guidelines. The wt-GIST cohort of LITESPARK-015 (NCT04924075) has not yet reported — watch ASCO 2026 / WCGIC for the interim readout. The cross-tumor mechanism rationale (SDH loss → HIF-2α stabilization, Selak 2005) is solid. · **Geographic scope:** US + EU + AU + UK + CA recruiting LITESPARK-015 sites. Approved in US and EU for non-GIST indications. · **Verified:** 2026-06-26

Two real paths. The LITESPARK-015 wt-GIST cohort is recruiting at 14 US sites — that is the cleanest entry point if the patient relapses with measurable disease. Off-label belzutifan post-PPGL-approval is technically possible but expensive and likely to draw a prior-auth fight; compassionate-use through Merck is the path if both trial enrollment and standard off-label fail.

Next steps

1. Resolve the SDHA/SDHB IHC discrepancy (retained SDHB staining is unusual given biallelic SDHA inactivation) — sponsor will not accept dSDH-GIST labeling without it.
2. Resolve the germline SDHA call to confirm constitutional vs sporadic SDH-deficient driver.
3. At measurable recurrence: email Trialsites@msd.com (or call 1-888-577-8839) to identify the nearest LITESPARK-015 site and request screening slot.
4. If trial-ineligible: submit an expanded-access request via Merck's compassionate-use portal citing the PPGL-cohort data as the closest mechanism-matched precedent.
5. Watch ASCO 2026 / WCGIC for the LITESPARK-015 wt-GIST cohort interim readout — could shift this to a stronger off-label case.

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT04924075	2	recruiting	likely	Toll Free Number (Merck Trial Sites); Trialsites@msd.com; 1-888-577-8839

Manufacturer / sponsor contact

- **Company:** Merck Sharp & Dohme LLC
- **Med info phone:**1-800-672-6372
- **Product info:**https://www.welireghcp.com/
- **Compassionate / expanded access:**https://www.merck.com/research/expanded-access/
- **Notes:** Merck Access Program for Welireg: 1-855-257-3932. The compassionate-use portal handles expanded-access requests for indications outside the current approvals — relevant here because GIST is not yet on the label.

Payer / coverage: Off-label for GIST. Payers will require documentation of the dSDH-GIST diagnosis, prior TKI failures, and a citation to the cross-tumor PPGL data (LITESPARK-015 PPGL cohort: ORR 26%, mPFS 22 mo). Expect a denial-appeal cycle. The Merck Patient Assistance Program may help for uninsured patients. Approximate WAC ~$25,000-30,000/month.

Notes: Belzutifan is the sole active HIF-2α option — DFF332 and NKT-2152 are both discontinued. Single point of failure if the wt-GIST cohort reads out negative.

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4. cobimetinib (Cotellic) — MEK inhibitor (MEGALiT arm) for MAP2K1 P124S

Aliases: cobimetinib, GDC-0973, Cotellic, XL518

Modality: small_molecule · **Regulatory:** FDA-approved (BRAF V600E/K melanoma in combination with vemurafenib; histiocytic neoplasms monotherapy 2022). EMA-approved. · **Guidelines:** Not on NCCN GIST guidelines. Same mechanism rationale as trametinib for MAP2K1 P124S — class I RAF-dependent activator, MEK-sensitive in vitro. No tumor-specific evidence in GIST. · **Geographic scope:** Trial: EU only (Sweden, closed). Off-label Rx: US + EU. · **Verified:** 2026-06-26

Practically interchangeable with trametinib for the MAP2K1 P124S question, but with weaker access — MEGALiT is closed and there is no active cobimetinib-specific MAP2K1 basket. If a MEK inhibitor enters the plan, trametinib is the better-supported choice. Cobimetinib stays in the dossier for completeness.

Next steps

1. Default to trametinib for MAP2K1 P124S off-label consideration unless a cobimetinib-specific MAP2K1 trial opens.
2. If MEGALiT reopens or a successor protocol with cobimetinib appears, confirm tissue MAP2K1 P124S and re-evaluate.

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT04185831	2	active not recruiting	no	Peter Nygren, MD, PhD (PI)

Manufacturer / sponsor contact

- **Company:** Genentech (Roche Group)
- **Med info phone:**1-800-821-8590
- **Product info:**https://www.gene.com/medical-professionals/medicines/cotellic
- **Notes:** Genentech Access Solutions: 1-877-436-3683 Mon-Fri 6am-5pm PT. Medical Information available via online form and chat in addition to phone.

Payer / coverage: Off-label for GIST and MAP2K1 alone. The histiocytic-neoplasm 2022 monotherapy approval does not extend to GIST. Approximate WAC ~$11,000/month. Same prior-auth challenges as trametinib.

Notes: Carried in the dossier because MEGALiT used cobimetinib specifically. Trametinib is the substitute the prescribing oncologist would actually use.

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5. imatinib (Gleevec) — adjuvant or first-line

Aliases: imatinib, imatinib mesylate, STI-571, Gleevec, Glivec

Modality: small_molecule · **Regulatory:** FDA-approved (KIT-positive GIST, adjuvant and unresectable/metastatic; CML and several other indications). EMA-approved. · **Guidelines:** ESMO-EURACAN-GENTURIS 2022 explicitly recommends against adjuvant imatinib in SDH-deficient GIST. NCCN lists imatinib as standard adjuvant only for KIT-mutant high-risk GIST; KIT/PDGFRA-WT subtypes including dSDH-GIST are flagged as imatinib-insensitive. · **Geographic scope:** Global; widely available as generic. · **Verified:** 2026-06-26

Imatinib is FDA-approved and easy to get on-label, but it does not belong in this patient's plan. Boikos 2016 showed a 1/49 (2%) PR rate in dSDH-GIST, and the ESMO guideline recommends against it. The off-label classification is technical — there is no rationale to use the off-label path here.

Next steps

1. Document the explicit decision against adjuvant imatinib in the chart citing Boikos 2016 (PMID 27011036) and the ESMO-EURACAN-GENTURIS guideline.
2. If the surgical team asks about 3-year adjuvant imatinib (PERSIST-5, SSGXVIII), note that those trials excluded the dSDH-GIST population structurally and the imatinib-insensitive genotypes recurred regardless of treatment duration.

Manufacturer / sponsor contact

- **Company:** Novartis Pharmaceuticals Corporation
- **Med info phone:**1-888-669-6682 (1-888-NOW-NOVA)
- **Product info:**https://www.novartis.com/us-en/content/gleevec%C2%AE-tablets
- **Notes:** Imatinib is widely available as generic; the Novartis branded Gleevec medical-info line is the right call only if a clinician needs label clarification.

Payer / coverage: On-label adjuvant imatinib is universally covered for KIT-mutant high-risk GIST. Off-label adjuvant use in KIT/PDGFRA-WT (including dSDH) GIST is typically NOT reimbursed by Medicare or commercial payers without prior auth and a strong biomarker case — which this patient does not have.

Notes: Included for completeness because the user asked. The drug is accessible; the rationale is not.

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6. inavolisib (Itovebi) — selective PI3Kα inhibitor/degrader for PIK3CA R93W

Aliases: inavolisib, GDC-0077, Itovebi

Modality: small_molecule · **Regulatory:** FDA-approved (HR+/HER2- PIK3CA-mutated advanced breast cancer with palbociclib + fulvestrant 2024). EMA-approved (2025). · **Guidelines:** NCCN-supported only for HR+/HER2- mBC with canonical PIK3CA hotspots per the registration trial population. Not on NCCN GIST guidelines. · **Geographic scope:** US + EU — recently approved in both markets for the HR+/HER2- mBC indication. · **Verified:** 2026-06-26

Same logic as alpelisib but with a cleaner mechanism (selective PI3Kα + mutant-allele degrader). The non-canonical R93W issue still applies. If a PI3Kα-pathway path is pursued, the tersolisib trial is the right experiment first; off-label inavolisib stays a fallback if approved/selective trials are exhausted.

Next steps

1. Tissue NGS confirmation of PIK3CA R93W (currently ctDNA-only) is the gate.
2. Default to the tersolisib trial (NCT05768139); off-label inavolisib has no advantage over alpelisib without comparative data in non-canonical variants.
3. If commercial-payer denial cycle becomes the question, the Genentech Access Solutions team handles the appeals (1-877-436-3683).

Manufacturer / sponsor contact

- **Company:** Genentech (Roche Group)
- **Med info phone:**1-800-821-8590
- **Product info:**https://www.itovebi.com/
- **Notes:** Genentech Access Solutions: 1-877-436-3683 Mon-Fri 6am-5pm PT. Newer drug than alpelisib with a kinase-selective + degrader mechanism that improves the therapeutic window in breast-cancer registration data.

Payer / coverage: Off-label for GIST and for non-canonical PIK3CA variants. The INAVO120 trial used canonical hotspots; payers will mirror the label. Approximate WAC similar to alpelisib (~$17,000-20,000/month).

Notes: Newer than alpelisib and mechanistically interesting, but the variant problem dominates. Off-label inavolisib for dSDH-GIST is a long shot.

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7. olaparib + temozolomide combination — PARP + alkylator for dSDH-GIST contingency

Aliases: olaparib, AZD2281, Lynparza, temozolomide, TMZ, Temodar

Modality: small_molecule · **Regulatory:** Both components individually FDA- and EMA-approved (olaparib: BRCA1/2-mutated breast/ovarian/pancreatic/prostate cancer, HRD-positive ovarian maintenance; temozolomide: glioblastoma, anaplastic astrocytoma, others). Combination is not approved. · **Guidelines:** Not on NCCN GIST guidelines. The mechanism rationale rests on Sulkowski 2018 (Nat Genet) + Sulkowski 2020 (Nature) — SDH/FH-loss creates a BRCA-like phenotype at chromatin via succinate-mediated KDM4B inhibition. Singh 2023 (PMID 36151992) is the single-patient case report showing clinical response in multiply-relapsed dSDH-GIST + PGL. · **Geographic scope:** US + EU + AU + JP — both drugs approved in major markets, off-label for this combination and this indication. · **Verified:** 2026-06-26

Mechanism-sound but evidence-thin. The Sulkowski 2018/2020 papers establish the BRCA-like phenotype in SDH-loss models, and Singh 2023 reports one durable response in a multiply-relapsed patient. With no trial currently enrolling this combination in dSDH-GIST and an aggressive prior-auth fight ahead, this is a late-line option after the temozolomide and pemigatinib trials are exhausted.

Next steps

1. Reserve for late-line use after sunitinib/regorafenib/ripretinib and after temozolomide and pemigatinib trial paths are exhausted.
2. If pursued: order MGMT methylation (response biomarker for the TMZ component) and HRD assay on tumor tissue.
3. Off-label prior-auth packet should cite Sulkowski 2018 (PMID 30013182), Sulkowski 2020 (PMID 32494005), and Singh 2023 (PMID 36151992).
4. AstraZeneca Access 360 (1-844-275-2360) for copay support if approved; Merck patient-assistance for temozolomide (1-800-727-5400) — note that generic TMZ keeps the alkylator cost low.

Manufacturer / sponsor contact

- **Company:** AstraZeneca (olaparib) + Merck (temozolomide)
- **Med info phone:**1-800-236-9933 (AstraZeneca, olaparib) / 1-877-888-4231 (Merck, temozolomide)
- **Product info:**https://www.lynparzahcp.com/
- **Compassionate / expanded access:**https://medicalinformation.astrazeneca-us.com/
- **Notes:** AstraZeneca Access 360 for olaparib: 1-844-275-2360 (1-844-ASK-A360) Mon-Fri 8am-6pm ET. AZ&Me Prescription Savings Program: 1-800-292-6363. Olaparib WAC ~$17,000/month.

Payer / coverage: Both components are off-label in this combination and in this indication. Expect categorical denial on first prior auth — the case rests on the Sulkowski mechanism papers and one case report. Cash-pay olaparib is prohibitive; AstraZeneca Access 360 patient-assistance is the realistic financial path if approved.

Notes: No active trial of this combination in dSDH-GIST. If a trial opens, that becomes the dominant path.

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8. pemigatinib (Pemazyre) — FGFR inhibitor for dSDH-GIST contingency

Aliases: pemigatinib, INCB054828, Pemazyre

Modality: small_molecule · **Regulatory:** FDA-approved (cholangiocarcinoma with FGFR2 fusion/rearrangement 2020; myeloid/lymphoid neoplasm with FGFR1 rearrangement 2022). EMA-approved (cholangiocarcinoma). · **Guidelines:** Not on NCCN GIST guidelines. The mechanism rationale rests on Flynn 2025 AACR abstract CT215 — significant tumor regression in an SDH-deficient GIST PDX and 41.7% ORR / 31-month mPFS in the parallel rogaratinib NCT04595747 trial in dSDH-GIST. · **Geographic scope:** US only — DFCI single-site enrollment. Approved in US/EU for cholangiocarcinoma. · **Verified:** 2026-06-26

Email Suzanne George at DFCI for the PEMIGIST trial slot if the patient relapses with measurable disease — that is the practical access path. Off-label pemigatinib outside trials is hard to justify because the patient does not carry a canonical FGFR alteration; the rationale rides on the FGF3/4 overexpression mechanism Flynn 2025 described, which is not a companion-diagnostic finding.

Next steps

1. Confirm SDH-deficient phenotype with SDHA/SDHB IHC re-stain and germline SDHA resolution.
2. At measurable recurrence, email Suzanne George (suzanne_george@dfci.harvard.edu) about a PEMIGIST slot.
3. Consider FGFR1/3 + FGF3/4 expression profiling on tumor tissue if available — this would strengthen both the trial case and any off-label appeal.
4. Watch for parallel readouts on the rogaratinib trial (NCT04595747) — could broaden the FGFR-inhibitor case in dSDH-GIST beyond pemigatinib.

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT07434843	2	recruiting	likely	Suzanne George, MD; suzanne_george@dfci.harvard.edu; 617-632-5204

Manufacturer / sponsor contact

- **Company:** Incyte Corporation
- **Med info phone:**1-855-463-3463
- **Product info:**https://www.pemazyre.com/
- **Compassionate / expanded access:** https://www.incyte.com/our-science/our-medicines/expanded-access-policy
- **Notes:** IncyteCARES patient-support line: 1-855-452-5234 Mon-Fri 8am-8pm ET. Incyte's expanded-access policy is published.

Payer / coverage: Off-label for GIST. Approximate WAC ~$20,000/month. Likely to require trial enrollment for access; off-label prior-auth without companion-diagnostic FGFR alteration will be difficult. IncyteCARES financial-assistance enrollment helps with copay if approved.

Notes: Strongest preclinical-to-clinic translation in dSDH-GIST so far per the Flynn 2025 abstract. Trial enrollment is the realistic path; off-label is a long shot without companion-diagnostic FGFR alterations.

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9. regorafenib (Stivarga) — 3L TKI for dSDH-GIST contingency

Aliases: regorafenib, BAY 73-4506, Stivarga

Modality: small_molecule · **Regulatory:** FDA-approved (advanced GIST after imatinib + sunitinib; metastatic

CRC; HCC after sorafenib). EMA-approved. · **Guidelines:** NCCN-supported as 3L TKI for advanced GIST. dSDH-GIST cohort data is by extrapolation; Dedousis 2025 review (PMID 40045030) calls activity 'modest, similar to KIT-mutant 3L'. · **Geographic scope:** Global; approved in US, EU, JP, and most major markets. · **Verified:** 2026-06-26

Standard 3L option after sunitinib failure. The dSDH-GIST evidence base is thinner than for sunitinib (GRID enrolled mostly KIT-mutant disease), but regorafenib is the accepted 3L per NCCN and is reimbursed without controversy. Toxicity counseling matters more than access here.

Next steps

1. On sunitinib progression, start regorafenib 160 mg PO daily 3 weeks on / 1 week off.
2. Pre-treatment: BP optimization, dental check, dermatology baseline for HFSR.
3. Enroll in Bayer's $0 copay program if commercially insured (1-866-581-4992).
4. Plan dose modifications proactively — most patients need to drop to 120 mg or alternate-week dosing within the first 2-3 cycles.

Manufacturer / sponsor contact

- **Company:** Bayer HealthCare Pharmaceuticals
- **Med info phone:**1-888-842-2937 (1-888-84-BAYER)
- **Product info:**https://www.stivarga-us.com/
- **Compassionate / expanded access:**https://www.bayer.com/en/pharma/expanded-access-policy
- **Notes:** Bayer Access Services: 1-800-288-8374 for insurance navigation. $0 copay program for commercially insured: 1-866-581-4992. Bayer US Patient Assistance Foundation for uninsured: 1-866-228-7723.

Payer / coverage: Advanced GIST is on-label after imatinib + sunitinib; Medicare and commercial coverage is routine. Hand-foot syndrome and hypertension management drive most of the practical complexity, not access.

Notes: Same off-label-vs-on-label note as sunitinib — the label covers advanced GIST without molecular stratification, so payers treat it as on-label.

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10. ripretinib (Qinlock) — 4L+ TKI for dSDH-GIST contingency

Aliases: ripretinib, DCC-2618, Qinlock

Modality: small_molecule · **Regulatory:** FDA-approved (advanced GIST after ≥3 prior TKIs including imatinib). EMA-approved. · **Guidelines:** NCCN-supported as 4L+ TKI for advanced GIST. Mavroeidis 2025 review notes the dSDH-GIST evidence base for ripretinib is the weakest of the three post-imatinib TKIs because INVICTUS enrolled vanishingly few wt-GIST patients; use is by extrapolation. · **Geographic scope:** US + EU + JP; approved in most major markets. · **Verified:** 2026-06-26

Fourth-line option once sunitinib and regorafenib are exhausted. The drug is FDA-approved and Deciphera's patient-access program (AccessPoint) is unusually robust — rapid-start, bridge coverage, and full free drug for uninsured. Practical access is rarely the barrier; the question is whether ripretinib outperforms simply rotating to a clinical trial at this stage in dSDH-GIST.

Next steps

1. Document prior imatinib, sunitinib, regorafenib lines in the chart before submitting prior auth (label requires ≥3).
2. Call Deciphera AccessPoint (1-833-432-2237) to enroll before the first prescription — they handle the prior-auth paperwork.
3. Re-evaluate the trial landscape at the 4L transition point — the temozolomide and pemigatinib SDH-deficient-GIST trials may be the better answer at recurrence than ripretinib, depending on what reads out by then.

Manufacturer / sponsor contact

- **Company:** Deciphera Pharmaceuticals
- **Med info phone:**1-888-724-3274
- **Med info email:**info@decipheraaccesspoint.com
- **Product info:**https://www.qinlockhcp.com/
- **Notes:** Deciphera AccessPoint: 1-833-432-2237 (1-833-4DACCES) Mon-Fri 8am-8pm ET. Programs include Rapid Start (insurance bridge), commercial copay support (as low as $0), and free drug for uninsured.

Payer / coverage: On-label after ≥3 prior TKIs. The order-of-prior-lines requirement is enforced by payers — make sure the patient has documented imatinib + sunitinib + regorafenib exposure (even if imatinib was a brief trial of futility) before requesting ripretinib coverage.

Notes: Standard of care for KIT-mutant 4L GIST. For dSDH-GIST the evidence is case-level only — Mavroeidis 2025 review is the right citation for the 'weakest of the three TKIs' framing.

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11. sunitinib (Sutent) — 2L TKI for dSDH-GIST contingency

Aliases: sunitinib, SU011248, Sutent

Modality: small_molecule · **Regulatory:** FDA-approved (advanced GIST after imatinib failure/intolerance; RCC; pNET; adjuvant RCC). EMA-approved. · **Guidelines:** NCCN-supported as 2L TKI in advanced GIST regardless of subtype; Boikos 2016 (NIH wt-GIST clinic) provides the dSDH-specific anchor — 18% ORR vs imatinib's 2%, the empirical reason sunitinib is the preferred first systemic option at recurrence. · **Geographic scope:** Global; approved in US, EU, AU, JP, and most major markets. · **Verified:** 2026-06-26

If the patient recurs, sunitinib 50 mg PO 4-on/2-off is the first-line TKI of choice on the strength of Boikos 2016 plus the von Mehren / Dedousis 2025 reviews. The drug is approved, widely stocked, and the prescription is straightforward — the practical question is timing (wait for measurable progression, then start) rather than access.

Next steps

1. At first measurable recurrence, start sunitinib 50 mg PO daily 4 weeks on / 2 weeks off. Confirm dosing with the prescribing oncologist; some dSDH-GIST patients tolerate 37.5 mg continuous better.
2. Baseline TSH, BP, LFTs, CBC; recheck cycles 1-3 then quarterly.
3. Pfizer Oncology Together: have the patient enroll up front for copay support (1-877-744-5675).
4. If progression on sunitinib, move to regorafenib (3L).

Manufacturer / sponsor contact

- **Company:** Pfizer Inc.

- **Med info phone:** 1-800-505-4426
- **Product info:** https://www.pfizermedical.com/patient/sutent
- **Compassionate / expanded access:** https://www.pfizer.com/science/clinical-trials/expanded-access
- **Notes:** Pfizer Oncology Together helps with insurance navigation and patient-assistance enrollment. Patient-assistance line: 1-877-744-5675.

Payer / coverage: GIST is an on-label indication; Medicare and commercial payers cover sunitinib for advanced GIST without molecular stratification. Prior auth is routine but uncomplicated. Pfizer Oncology Together copay-assistance program available for eligible commercially insured patients.

Notes: Technically off-label in the sense that the GIST label is anchored on KIT-mutant cohorts and the dSDH subset was barely represented in the registration trial — but no payer treats this as off-label; reimbursement is routine.

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12. temozolomide (Temodar) — targeted off-label or trial-based for dSDH-GIST

Aliases: temozolomide, TMZ, Temodar, Temodal

Modality: small_molecule · **Regulatory:** FDA-approved (newly diagnosed glioblastoma with radiotherapy and as maintenance; anaplastic astrocytoma; refractory anaplastic astrocytoma — also approved 2024 for additional indications under Project Renewal). EMA-approved. Widely available as generic. · **Guidelines:** Not on NCCN GIST guidelines. Off-guideline but supported by Yebra 2022 (PMID 34426440; 2/5 PR, 100% DCR in dSDH-GIST mini-cohort) plus the parallel preclinical PDX data. MGMT promoter methylation (Flego 2024 PMID 38132569, Giger 2022 PMID 36198483) is the candidate response biomarker. · **Geographic scope:** Trial: Korea-only currently. Off-label Rx: US + EU + AU + JP — temozolomide is approved and generic everywhere. · **Verified:** 2026-06-26

Two access paths here. Trial-only via the Asan/Korea trial (NCT05661643) is the cleanest if the patient can travel; UCSD's NCT03556384 is no longer enrolling. Off-label TMZ via generic Rx is the practical fallback at first recurrence — supported by Yebra 2022 and the MGMT-methylation rationale, but expect a prior-auth fight with most US payers.

Next steps

1. At first recurrence with measurable disease, order MGMT promoter methylation on tumor tissue (or re-cut from the resection block) — this is the load-bearing biomarker for off-label approval and trial enrollment.
2. If MGMT-methylated and travel-capable: email Min-Hee Ryu (miniryu@amc.seoul.kr) about an Asan slot in NCT05661643.
3. If not travel-capable: prepare an off-label prior-auth packet with the Yebra 2022 PMID, MGMT result, dSDH diagnosis, and the failed-imatinib/sunitinib lines once those occur. Generic TMZ keeps cash-pay viable.
4. Watch for the NCT03556384 readout (UCSD) — primary completion June 2025 per the registry; data could change the off-label-prior-auth calculus.

Trial pathways

NCT	Phase	Status	Eligible	Central contact

NCT05661643	recruiting	likely	Min-Hee Ryu, MD, PhD; miniryu@amc.seoul.kr; 82-2-3010-5936
NCT03556384	active not recruiting	no	—

Manufacturer / sponsor contact

- **Company:** Merck Sharp & Dohme LLC (branded Temodar); multiple generic manufacturers
- **Med info phone:**1-877-888-4231
- **Product info:**https://www.merck.com/product/usa/pi_circulars/t/temodar_capsules/temodar_pi.pdf
- **Notes:** Merck Patient Assistance Program (Temodar): 1-800-727-5400. Generic temozolomide is the typical prescription — much lower out-of-pocket cost than branded Temodar.

Payer / coverage: Off-label for GIST. Generic availability means cash-pay is feasible if prior auth fails. Payers often approve off-label TMZ when the chart shows MGMT methylation, dSDH-GIST molecular phenotype, and a citation to Yebra 2022 — request the off-label review with these elements documented up front.

Notes: MGMT methylation is the response-enrichment lever. Without it, off-label TMZ becomes harder to justify and the Asan trial inclusion is on weaker ground.

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13. trametinib (Mekinist) — MEK inhibitor for MAP2K1 P124S

Aliases: trametinib, GSK1120212, Mekinist

Modality: small_molecule · **Regulatory:** FDA-approved (BRAF V600E/K melanoma in combination with dabrafenib; BRAF V600E NSCLC; BRAF V600E anaplastic thyroid; BRAF V600E pediatric low-grade glioma). EMA-approved. · **Guidelines:** Not on NCCN GIST guidelines. The MAP2K1 P124S rationale rests on Gao 2018 functional characterization (RAF-dependent class I activator, MEK-inhibitor sensitive in Ba/F3) plus the Gounder 2018 NEJM histiocytic sarcoma case report (MAP2K1 F53L → durable CR on trametinib). Cross-tumor extrapolation. · **Geographic scope:** Trial: China (NCT06739395 only currently enrolling). Off-label Rx: US + EU + JP. Trametinib is approved in major markets but only for BRAF-mutant indications. · **Verified:** 2026-06-26

Two real questions before this becomes actionable. First: is the MAP2K1 P124S real and clonal? It is ctDNA-only right now and tissue confirmation is gated by FoundationOne CDx re-test. Second: trial vs off-label. The Tianjin trial (NCT06739395) is geographically blocked for most patients, MEGALiT looks closed, so off-label trametinib via Novartis is the practical path if the variant confirms — expect a prior-auth fight.

Next steps

1. Order FoundationOne CDx tissue NGS (or paired tumor + normal) to confirm MAP2K1 P124S as clonal in tumor rather than ctDNA noise.
2. If confirmed clonal: prepare an off-label prior-auth packet citing Gao 2018 (PMID 36442478), Gounder 2018 (PMID 26222557), and the Wagle 2014 P124 αC-helix mechanism (PMID 24265154).
3. Confirm no new MAP2K1 basket trial has opened in the US/EU since the MEGALiT/MyCustom-02 cohorts — recheck quarterly.
4. If the trial path is the patient's priority and US travel-only: monitor MATCH-II / NCI-MATCH-style successor protocols for a MAP2K1 arm.

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT06739395	2	recruiting	unconfirmed	Haitao Wang, Ph.D; peterrock2000@126.com; +86-022-88326385
NCT04185831	2	active not recruiting	no	Peter Nygren, MD, PhD (PI)

Manufacturer / sponsor contact

- **Company:** Novartis Pharmaceuticals Corporation
- **Med info phone:**1-888-669-6682 (1-888-NOW-NOVA)
- **Product info:**https://www.novartis.com/us-en/content/mekinist%C2%AE-tablets
- **Notes:** Novartis Patient Support: 1-800-282-7630 Mon-Fri 8am-8pm ET. Novartis Oncology Universal Co-pay Program: 1-877-577-7756.

Payer / coverage: Off-label for GIST and for MAP2K1-only indications. Approximate WAC ~$13,000/month for monotherapy. Off-label prior auth needs tissue-confirmed MAP2K1 P124S plus a citation to Gao 2018 and Gounder 2018; payers are inconsistent and often deny on first pass. NCCN's tumor-agnostic Compendium does not list trametinib for MAP2K1 alone.

Notes: P124S is the αC-helix variant Wagle 2014 characterized in melanoma as a partial-resistance / partial-activating mutation. Cross-tumor + cross-class — the case for monotherapy MEK in dSDH-GIST is real but not strong. Tissue confirmation is the gate.

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14. vandetanib (Caprelsa) — VEGFR/RET/EGFR TKI

Aliases: vandetanib, ZD6474, Caprelsa

Modality: small_molecule · **Regulatory:** FDA-approved (Caprelsa) for symptomatic or progressive medullary thyroid cancer that is unresectable, locally advanced, or metastatic. EMA-approved. FDA removed the Caprelsa REMS in 2025. · **Guidelines:** Not on GIST guidelines. The dSDH-GIST trial was negative: Glod and Widemann 2019 (PMID 31439578, NCT02015065) saw 0/9 objective responses, with treatment-limiting toxicity in 3 of 5 adults dosed at 300 mg. · **Geographic scope:** US + EU; approved for medullary thyroid cancer in both. · **Verified:** 2026-06-26

The drug is FDA-approved, so off-label prescription is technically on the table, but there is no reason to use it here. The NCI trial in exactly this molecular subtype returned zero responses and notable adult toxicity. This row exists to record that the VEGFR/RET-directed single-agent idea was tested and did not work.

Next steps

1. Do not pursue off-label vandetanib for this patient; the dSDH-GIST trial was a hard negative.
2. If the surgical or oncology team raises it, cite Glod/Widemann 2019 (PMID 31439578) and point to sunitinib as the better-supported VEGFR-active TKI at recurrence.

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT02015065	2	completed	no	—

Manufacturer / sponsor contact

- **Company:** Sanofi (Genzyme)
- **Med info phone:**1-800-633-1610
- **Product info:**https://www.sanofi.us/en/your-health/products/prescription-products/caprelsa
- **Notes:** 1-800-633-1610 is the Sanofi Genzyme medical-information and adverse-event line. Caprelsa is the branded vandetanib product.

Payer / coverage: On-label and covered for medullary thyroid cancer. Off-label for GIST would draw a prior-auth denial, and there is no efficacy case to mount given the negative dSDH-GIST trial.

Notes: Included because the user asked. Approved drug, off-label path open in principle, no rationale to use it.

Clinical trial (5)

15. NCI Rare Tumor Natural History Study (registry + specialty referral)

Aliases: NCI Rare Tumor Natural History Study, NCI Pediatric Oncology Branch rare-tumor protocol

Modality: other · **Regulatory:** n/a — observational registry, no investigational drug. · **Guidelines:** n/a — entry point into the NCI Pediatric & wt-GIST Clinic referral network. · **Geographic scope:** US — NIH Clinical Center (Bethesda, MD), OHSU (Portland, OR), Texas Children's Hospital (Houston, TX). · **Verified:** 2026-06-26

The cleanest specialty-care path for this patient right now. The NCI Pediatric & wt-GIST Clinic is the reference center for SDH-deficient GIST, and registry enrollment opens the door to SDHC-methylation testing, longitudinal biospecimen banking, and specialist follow-up alongside community surveillance. No drug treatment is administered through the protocol — it is the gateway, not the destination.

Next steps

1. Email Barbara Thomas (barbara.thomas2@nih.gov) or call 240-858-3633 to start enrollment.
2. Plan one in-person visit to NIH Bethesda for the baseline workup; subsequent imaging and labs can usually be shared from community care.
3. Bring the SDHA L306P / Y408N / E350fs constellation, the SDHB IHC result, and the resection pathology so the NCI clinic can run SDHC methylation and complete the molecular subtyping.
4. Couple registry enrollment with PGL screening and genetic counseling (Carney-Stratakis risk).

Trial pathways

NCT	Phase	Status	Eligible	Central contact

NCT03739827	n/a	recruiting	yes	Barbara J Thomas, R.N.; barbara.thomas2@nih.gov; (240) 858-3633

Manufacturer / sponsor contact

- **Company:** National Cancer Institute (Pediatric Oncology Branch)
- **Notes:** Federal sponsor; no manufacturer in the commercial sense.

Payer / coverage: Federally funded protocol; participation costs (including travel for some visits) often covered. Verify travel-support eligibility with the registry coordinator at enrollment.

Notes: This is a low-cost, high-yield action item for the case. Run it in parallel with active surveillance, not as a replacement for it.

16. RLY-2608 (zovegalisib) — mutant-selective PI3Kα inhibitor for PIK3CA R93W

Aliases: RLY-2608, RLY2608, zovegalisib

Modality: small_molecule · **Regulatory:** Investigational; no approval anywhere. Phase 1 in advanced solid tumors, advancing toward later-phase study in PIK3CA-mutant breast cancer. · **Guidelines:** Off-guideline; investigational. · **Geographic scope:** US + EU + AU recruiting sites. · **Verified:** 2026-06-26

Trial-only, and the most forgiving of the PI3Kα baskets for this patient's situation: it reads ctDNA, has an 'other solid tumors' group, and the sponsor can approve non-canonical variants one by one. The catch is the variant itself. R93W sits outside the hotspots these mutant-selective drugs are designed around, so the real gate is getting Relay to confirm it qualifies, on top of tissue-confirming the call. Lowest-confidence of the PIK3CA rows alongside tersolisib.

Next steps

1. Order tissue NGS (FoundationOne CDx or paired tumor/normal) to confirm PIK3CA R93W is clonal in tumor rather than ctDNA noise.
2. Email ClinicalTrials@relaytx.com (or call 617-322-0731) to ask whether PIK3CA R93W is on the trial's accepted variant list; this is the load-bearing question before any screening visit.
3. If accepted, find the nearest recruiting site through the registry sites list and request a slot at first measurable recurrence.
4. If not accepted, drop RLY-2608; there is no compassionate route and the variant is too weak to justify off-label PI3Kα inhibition.

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT05216432	1	recruiting	unconfirmed	Relay Therapeutics Inc; ClinicalTrials@relaytx.com; 617-322-0731

Manufacturer / sponsor contact

- **Company:** Relay Therapeutics, Inc.
- **Med info phone:**617-322-0731
- **Med info email:**ClinicalTrials@relaytx.com
- **Product info:**https://relaytx.com/patients/
- **Notes:** Relay states it does not run an expanded-access program and considers trial enrollment the appropriate route given the program's early stage. So there is no compassionate-use fallback; the trial is the only door.

Payer / coverage: Investigational and funded under the trial protocol, so no payer coverage applies.

Notes: RLY-2608 is also called zovegalisib. The eligibility language is the most likely to accommodate R93W if tissue confirms it, which is the only reason it stays in the dossier over alpelisib/inavolisib.

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17. olverembatinib (HQP1351) — multikinase inhibitor for dSDH-GIST contingency

Aliases: olverembatinib, HQP1351, GZD824, D824,

Modality: small_molecule · **Regulatory:** NMPA-approved in China (TKI-resistant T315I chronic- or accelerated-phase CML), and used in China for Ph+ ALL per CSCO guidance. Investigational for GIST everywhere; no FDA or EMA approval for any indication. · **Guidelines:** Not on NCCN or ESMO GIST guidelines. The dSDH-GIST signal comes from Qiu 2025 (PMID 41184234): 23.1% ORR and 25.7-month median PFS in a 26-patient SDH-deficient cohort, which is what the phase 3 NCT06640361 is built on. · **Geographic scope:** Approved in China only (CML / Ph+ ALL). The dSDH-GIST phase 3 is single-site in Guangzhou, China. No US or EU access outside a case-by-case named-patient request. · **Verified:** 2026-06-26

There is no clean US route to this drug right now, which is the honest headline despite it having the best dSDH-GIST efficacy data of the newer agents. The phase 3 sits at a single Guangzhou site, and the named-patient program is geared toward the approved leukemia indications rather than GIST. If the patient recurs, the realistic moves are to ask Ascentage whether named-patient GIST supply is possible and, separately, whether the China trial is feasible.

Next steps

1. Treat olverembatinib as a watch item, not an actionable option, while the patient is NED in the US.
2. If the patient recurs with measurable disease, resolve the SDHB-IHC re-stain first, since SDHB-IHC loss is the trial's entry biomarker and her current stain reads intact.
3. Email NPP-olverembatinib@ascentage.com to ask whether named-patient supply is available for SDH-deficient GIST, and on what terms, in the US.
4. Only if the patient is willing and able to travel: contact Yifan Zhai (yzhai@ascentage.com, +86-20-28068501) about the Guangzhou phase 3.
5. Track the phase 3 readout and any move toward a US or EU filing, which would change this row entirely.

Trial pathways

NCT	Phase	Status	Eligible	Central contact

NCT06640361	recruiting	no	Yifan Zhai, M.D., Ph.D.; yzhai@ascentage.com; +86-20-28068501

Manufacturer / sponsor contact

- **Company:** Ascentage Pharma Group Inc.
- **Product info:**https://www.ascentage.com/medical-information/patient-access-program/
- **Compassionate / expanded access:**https://www.ascentage.com/medical-information/expanded-access-policy/
- **Compassionate use email:**NPP-olverembatinib@ascentage.com
- **Notes:** Ascentage runs a named-patient program for olverembatinib (NPP-olverembatinib@ascentage.com) that supplies the drug where it is approved in one country but not the patient's. That program is built around the approved CML and Ph+ ALL indications; GIST is listed as investigational, so a GIST request would be case-by-case at the treating physician's initiative and is not guaranteed.

Payer / coverage: Not commercially available in the US, so no Medicare or commercial coverage applies. Any named-patient supply would be out-of-pocket and arranged drug-by-drug through Ascentage.

Notes: Strongest of the new dSDH-GIST agents on paper (Qiu 2025), weakest on US access. The mechanism story is non-KIT: VEGFR2/FGFR1/PDGFR/SRC blockade plus suppression of a HIF-driven CD36 lipid-scavenging program.

18. rogaratinib (BAY 1163877) — pan-FGFR inhibitor for dSDH-GIST

Aliases: rogaratinib, BAY 1163877, BAY1163877

Modality: small_molecule · **Regulatory:** Investigational; no FDA or EMA approval. Bayer has stepped back from the broader rogaratinib program, and the surviving study is the NCI sarcoma / dSDH-GIST phase 2. · **Guidelines:** Not on GIST guidelines, but this is the strongest dSDH-GIST readout to date: Merriam 2026 (PMID 42191879, Nature Medicine) reported 41.7% ORR and 31-month median PFS in the 24-patient SDH-deficient cohort of NCT04595747. · **Geographic scope:** US — 18 NCI-network sites ran the study, but it is closed to new enrollment. · **Verified:** 2026-06-26

Frustrating one. Rogaratinib has the best dSDH-GIST numbers of any systemic agent so far, but the only trial is closed to accrual, Bayer has stepped back, and there is no approval or compassionate route. The live way to test the same FGFR idea is the recruiting pemigatinib trial (NCT07434843) at DFCI, which is why that row, not this one, is the actionable FGFR path.

Next steps

1. Use this row as the evidence anchor for the FGFR-inhibitor rationale, not as an enrollment route, since NCT04595747 is closed to new patients.
2. For a live FGFR option at recurrence, go to the pemigatinib trial (NCT07434843) and contact Suzanne George at DFCI.
3. Ask the dSDH-GIST trial network whether a rogaratinib successor or expansion is planned given the positive Merriam 2026 readout.
4. Watch for any FDA filing or new sponsor for rogaratinib, which would reopen this path.

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT04595747	2	active not recruiting	no	—

Manufacturer / sponsor contact

- **Company:** Bayer (originator); National Cancer Institute / CTEP (sponsor and drug supply for the surviving trial)
- **Med info phone:** 1-888-842-2937
- **Notes:** Bayer is no longer actively developing rogaratinib, so the drug now lives inside the NCI trial rather than a commercial program. The Bayer medical-information line is useful only for legacy compound questions, not for access.

Payer / coverage: Investigational and supplied under the trial protocol, so no payer coverage applies. There is no manufacturer compassionate-use program now that Bayer has wound the program down.

Notes: Positive readout (ORR 41.7%) underwrites the whole FGFR-pathway case in dSDH-GIST (Flavahan 2019 insulator/FGF4 mechanism, Flynn 2025 PDX). The clinical baton has effectively passed to pemigatinib for new patients.

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19. tersolisib (STX-478) — selective PI3Kα inhibitor for PIK3CA R93W

Aliases: tersolisib, STX-478

Modality: small_molecule · **Regulatory:** Investigational; no regulatory approval anywhere. IND-cleared phase 1/2. · **Guidelines:** Off-guideline; investigational. · **Geographic scope:** US + EU + Japan recruiting sites. · **Verified:** 2026-06-26

Investigational trial-only path. The patient's PIK3CA R93W sits outside the validated alpelisib companion-diagnostic hotspots — Burke 2012 biochemistry shows ABD-domain variants like R88/R93 region are weakly activating, not strong drivers. Lilly's sponsor team will need to confirm R93W is on the trial's variant-eligibility list before screening; even with that, this is the lowest-confidence row in the dossier.

Next steps

1. Order FoundationOne CDx tissue NGS to confirm PIK3CA R93W as clonal in tumor (currently ctDNA-only).
2. Email LillyTrials@Lilly.com or call 1-877-285-4559 to ask whether PIK3CA R93W is on the trial's accepted variant list — this is the load-bearing question before any screening visit.
3. If accepted: find the nearest US site through the ClinicalTrials.gov sites URL and request a screening slot at first measurable recurrence.
4. If not accepted: drop tersolisib from the actionable plan; the variant is too weak to justify off-label PI3Kα inhibitor cash-pay.

Trial pathways

NCT	Phase	Status	Eligible	Central contact

NCT05768139	1/2	recruiting	unconfirmed	Eli Lilly Clinical Inquiry Hub; LillyTrials@Lilly.com; 1-877-285-4559 (1-877-CTLILLY)

Manufacturer / sponsor contact

- **Company:** Eli Lilly and Company (Scorpion Therapeutics acquired 2025)
- **Med info phone:**1-877-285-4559 (1-877-CTLILLY)
- **Med info email:**LillyTrials@Lilly.com
- **Product info:**https://www.lilly.com/clinical-trials
- **Notes:** STX-478 was Scorpion Therapeutics' lead asset; Lilly acquired the program in 2025 and now runs the trial.

Payer / coverage: Investigational; sponsor-funded under the trial protocol. No payer coverage applies.

Notes: Selective PI3Kα inhibitor with a wider therapeutic window than alpelisib in principle. The variant-eligibility question is the real gate.

Not yet accessible (2)

20. Comprehensive Multinational GIST Registry (observational)

Aliases: Comprehensive Multinational GIST Registry

Modality: other · **Regulatory:** Not applicable; observational registry with no investigational drug. · **Guidelines:** Not applicable. · **Geographic scope:** Germany (University Hospital Essen); not yet open. · **Verified:** 2026-06-26

This is a connective resource, not a treatment. Once it opens, the registry feeds a multinational GIST cohort the team can use to find trials and benchmark outcomes for a subtype with thin natural-history data. It is not yet recruiting and runs from Essen, so revisit it later rather than acting on it now.

Next steps

1. Treat as a bridge resource to check back on once it opens recruitment.
2. When it opens, email Johanna Falkenhorst (johanna.falkenhorst@uk-essen.de) to ask about remote or international participation, since the only listed site is in Germany.
3. Lean on the NCI Rare Tumor Natural History Study (NCT03739827) for registry/biobank participation in the meantime, which is already recruiting in the US.

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT07273045	n/a	not yet recruiting	likely	Johanna Falkenhorst, MD; johanna.falkenhorst@uk-essen.de; +40 201 723 2011

Manufacturer / sponsor contact

- **Company:** Universität Duisburg-Essen
- **Notes:** Academic registry sponsor; no manufacturer in the commercial sense.

Payer / coverage: Not applicable; observational registry.

Notes: New observational entry in the refreshed dossier. Value is data and trial-finding, not drug access.

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21. Wild-Type GIST Oncogenetic Panel Registry (observational)

Aliases: Wild Type GIST Oncogenetic Panel Registry

Modality: other · **Regulatory:** Not applicable; observational genomic-profiling registry with no investigational drug. · **Guidelines:** Not applicable. · **Geographic scope:** Taiwan; closed to new accrual. · **Verified:** 2026-06-26

Closed to new accrual and Taiwan-only, so for this patient it is a reference on the wt-GIST molecular landscape rather than something she enrolls in. Keep it in the picture for completeness of the registry options, but the NCI rare-tumor study is the practical one.

Next steps

1. Do not plan around this registry; it is closed to accrual and based in Taiwan.
2. Use it as published-data context on the wt-GIST genomic landscape if the team wants subtype benchmarking.

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT04821895	n/a	active not recruiting	no	—

Manufacturer / sponsor contact

- **Company:** National Health Research Institutes, Taiwan
- **Notes:** Academic registry sponsor; no manufacturer.

Payer / coverage: Not applicable; observational registry.

Notes: New observational entry in the refreshed dossier, retained for completeness of the wt-GIST registry picture.

Unavailable (4)

22. DFF332 — next-generation HIF-2α inhibitor

Aliases: DFF332

Modality: small_molecule · **Regulatory:** Investigational; never approved. Phase 1 terminated Feb 2026 for business reasons. · **Guidelines:** Off-guideline; discontinued. · **Geographic scope:** n/a — drug discontinued globally. · **Verified:** 2026-06-26

Closed door. The planned SDHx-mutation expansion arm never opened before Novartis terminated the program. Belzutifan is now the sole active HIF-2α inhibitor in development (NKT-2152 also discontinued 2025-2026). Kept in the dossier so the board sees the HIF-2α pipeline has been thinned to a single asset.

Next steps

1. Mark next-generation HIF-2α inhibitors as off the table beyond belzutifan.
2. Watch for new HIF-2α entrants but do not plan around them — the LITESPARK-015 wt-GIST cohort is the only HIF-2α path left.

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT04895748		terminated	no	—

Manufacturer / sponsor contact

- **Company:** Novartis Pharmaceuticals Corporation
- **Notes:** Development discontinued. No compassionate-access path.

Payer / coverage: n/a — drug not commercially available.

Notes: Audit-trail row. The implication for the case is that belzutifan is now a single-point-of-failure dependency in the HIF-2α strategy.

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23. guadecitabine (SGI-110) — DNA hypomethylating agent

Aliases: guadecitabine, SGI-110, S110

Modality: small_molecule · **Regulatory:** Investigational; never approved. Phase 3 trials in AML and MDS failed primary endpoints; development discontinued. · **Guidelines:** Off-guideline; discontinued program. · **Geographic scope:** n/a — drug discontinued globally. · **Verified:** 2026-06-26

Closed door. The phase 2 in wt-GIST/PPGL/HLRCC terminated for low accrual after 9 patients with zero objective responses, and the parent program was discontinued after phase 3 failures in AML/MDS. The hypomethylating-agent mechanism story is real for dSDH-GIST but the clinical readout was negative, and there is no compassionate-use path because Astex/Otsuka no longer supplies the drug.

Next steps

1. Mark hypomethylating-agent strategy as off the table for this patient.
2. If the methylator-phenotype story re-enters the discussion at recurrence (Killian 2013/2014 mechanism), pivot to other modalities — pemigatinib (FGFR), belzutifan (HIF-2α), or temozolomide (alkylator + MGMT

methylation).
3. Do not pursue decitabine/azacitidine as substitutes without a trial — the mechanism rationale that motivated guadecitabine was tested and read out negative.

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT03165721	2	terminated	no	—

Manufacturer / sponsor contact

- **Company:** Astex Pharmaceuticals (originally) / Otsuka Pharmaceuticals
- **Notes:** Compound development was suspended after the ASTRAL-1/2/3 phase 3 failures (~2019-2020). No active access path.

Payer / coverage: n/a — drug not commercially available.

Notes: Kept in the dossier so the board sees the hypomethylating-agent strategy was tested and is no longer pursuable.

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24. linsitinib (OSI-906) — IGF-1R/IR inhibitor

Aliases: linsitinib, OSI-906

Modality: small_molecule · **Regulatory:** Investigational; never approved. Astellas terminated development after multiple negative trials (adrenocortical carcinoma, GIST). · **Guidelines:** Off-guideline; discontinued. · **Geographic scope:** n/a — drug discontinued globally. · **Verified:** 2026-06-26

Closed door. SARC's phase 2 returned 0/20 RECIST responses and the parent program was shelved. The IGF-1R hypothesis in dSDH-GIST has been tested and is off the table; kept in the dossier as the audit trail for why the board does not need to revisit it.

Next steps

1. Mark IGF-1R inhibition as off the table.
2. If a next-generation IGF-1R agent ever re-enters trials, re-evaluate the hypothesis — but the dSDH-GIST clinical signal was a hard negative.

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT01560260	2	completed	no	—

Manufacturer / sponsor contact

- **Company:** Astellas Pharma (originally OSI Pharmaceuticals)
- **Notes:** Development discontinued after the adrenocortical and GIST trials read out negative. No current supply

or compassionate-access path.

Payer / coverage: n/a — drug not commercially available.

Notes: Same audit-trail role as guadecitabine — included so the board sees the hypothesis was tested.

25. telaglenastat (CB-839) — glutaminase inhibitor

Aliases: telaglenastat, CB-839, CB839

Modality: small_molecule · **Regulatory:** Investigational; never approved. Development discontinued and the asset divested. · **Guidelines:** Off-guideline; discontinued program. · **Geographic scope:** Not applicable; discontinued globally. · **Verified:** 2026-06-26

Closed door. Calithera wound the company down and sold off telaglenastat after its kidney and lung trials failed, and the SDH-deficient GIST cohort never read out positive. The glutaminase rationale also took a mechanistic hit from Cardaci 2015, which showed pyruvate carboxylation, not glutamine, carries most of the anaplerotic load when SDH is lost.

Next steps

1. Mark glutaminase inhibition as off the table for this patient.
2. If a metabolic-vulnerability strategy comes up again at recurrence, look at it fresh rather than reviving CB-839, which has no supply and a weak mechanistic case here.

Trial pathways

NCT	Phase	Status	Eligible	Central contact
NCT02071862		completed	no	—

Manufacturer / sponsor contact

- **Company:** Calithera Biosciences (dissolved 2023)
- **Notes:** Calithera's board approved dissolution and liquidation in January 2023 and sold the telaglenastat assets to a third party in March 2023, after the renal (ENTRATA/CANTATA) and lung (KEAPSAKE) trials missed. There is no compassionate-access path.

Payer / coverage: Not applicable; the drug is not commercially available.

Notes: Kept so the board sees the glutaminase-metabolism route was tried in this subtype and is no longer pursuable.